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(54) **FRICTION STIR WELDING APPARATUS AND FRICTION STIR WELDING METHOD**

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CPC ***B23K 20/1255*** (2013.01)

(57) **ABSTRACT**

A friction stir welding apparatus includes an anvil, a probe, and a shoulder member. The probe includes: a joining probe which advances toward plate members of a stacked body, forms a joint hole in the plate members softened by frictional heat generated by a joining surface of the joining probe coming into sliding contact with the plate members and joins the plate members, and forms an overlay portion protruding in a bank shape from a plate surface around the joint hole along an inner peripheral surface of the shoulder member; and a backfilling probe which advances toward the plate members, and backfills the joint hole with the overlay portion softened by frictional heat generated by a backfilling surface of the backfilling probe coming into sliding contact with the overlay portion. The joining probe and the backfilling probe are disposed to be replaceable with each other.

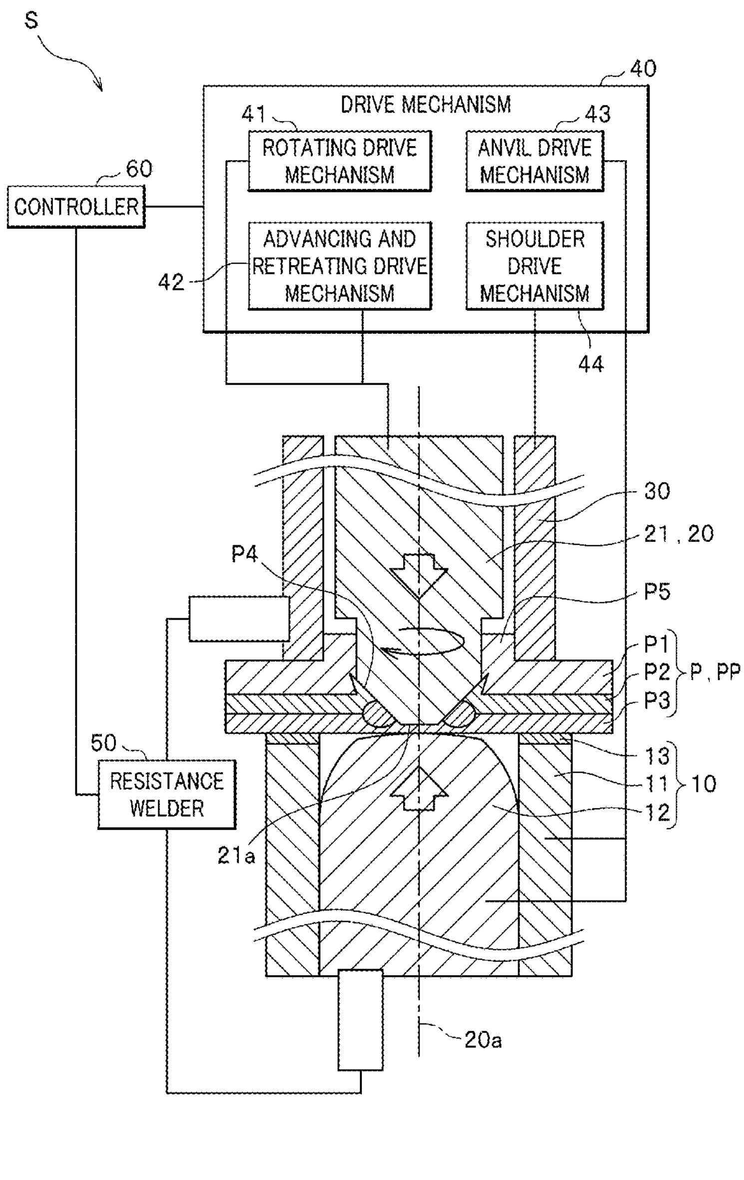


FIG. 1

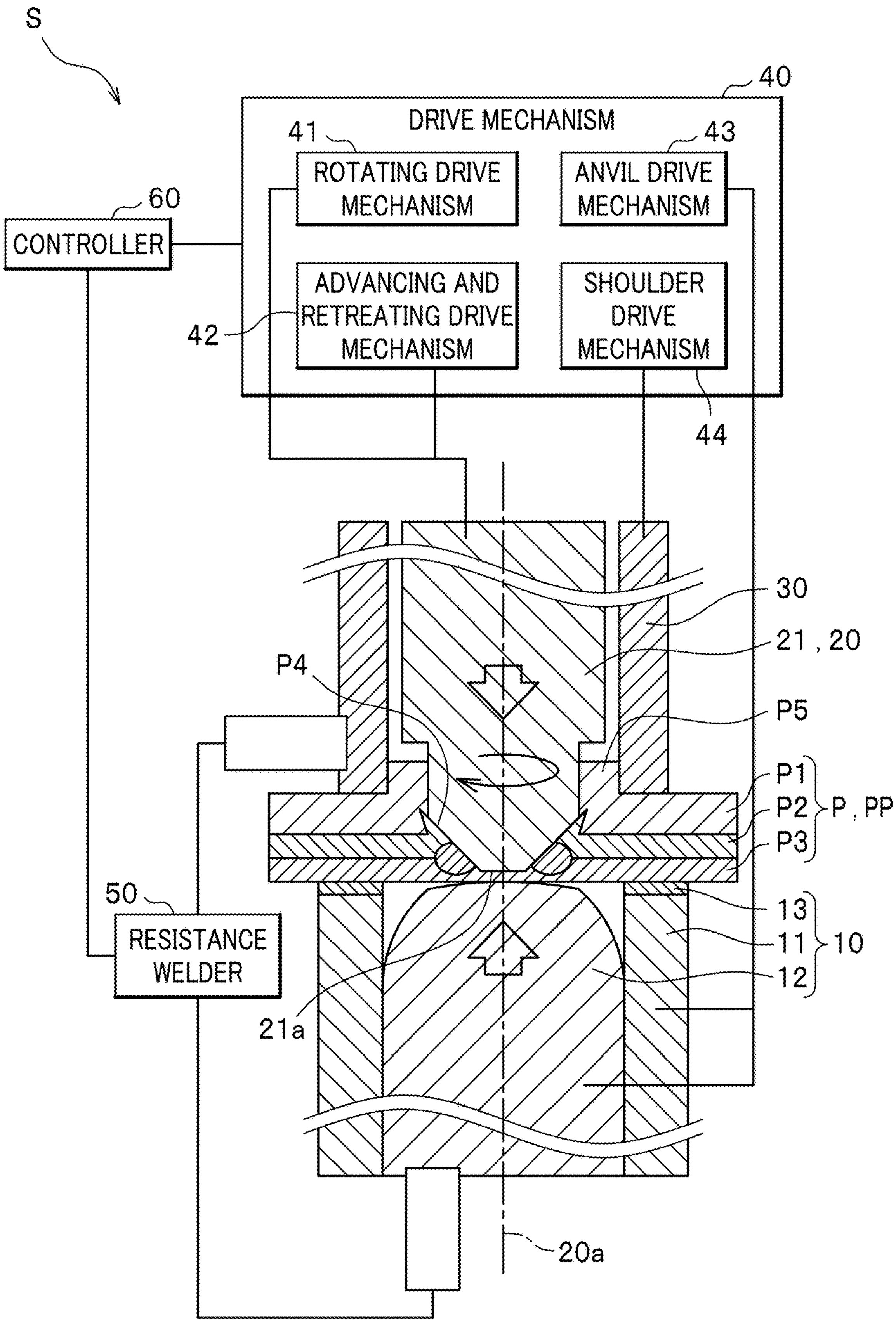

S
40
DRIVE MECHANISM
41
ROTATING DRIVE MECHANISM
43
ANVIL DRIVE MECHANISM
60
CONTROLLER
ADVANCING AND RETREATING DRIVE MECHANISM
42
SHOULDER DRIVE MECHANISM
44
30
21 , 20
P4
P5
P1
P2
P3
P , PP
13
11
10
12
50
RESISTANCE WELDER
21a
20a

FIG. 2

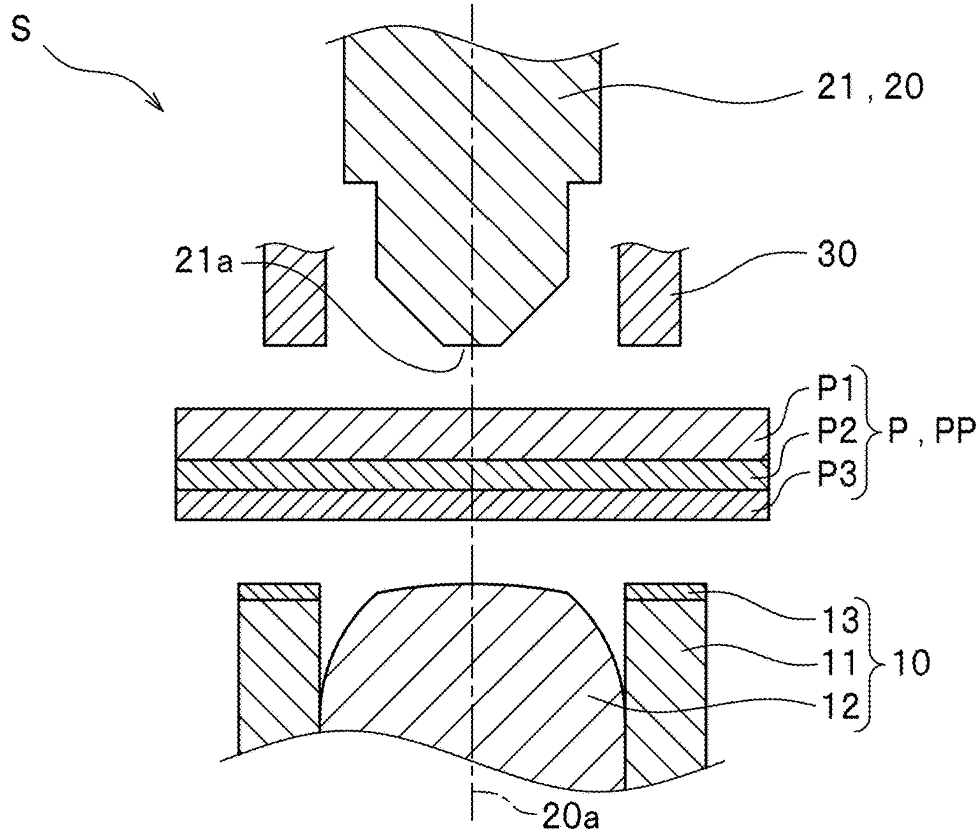

S
21, 20
21a
30
P1
P2
P3
P, PP
13
11
12
10
20a

FIG. 3

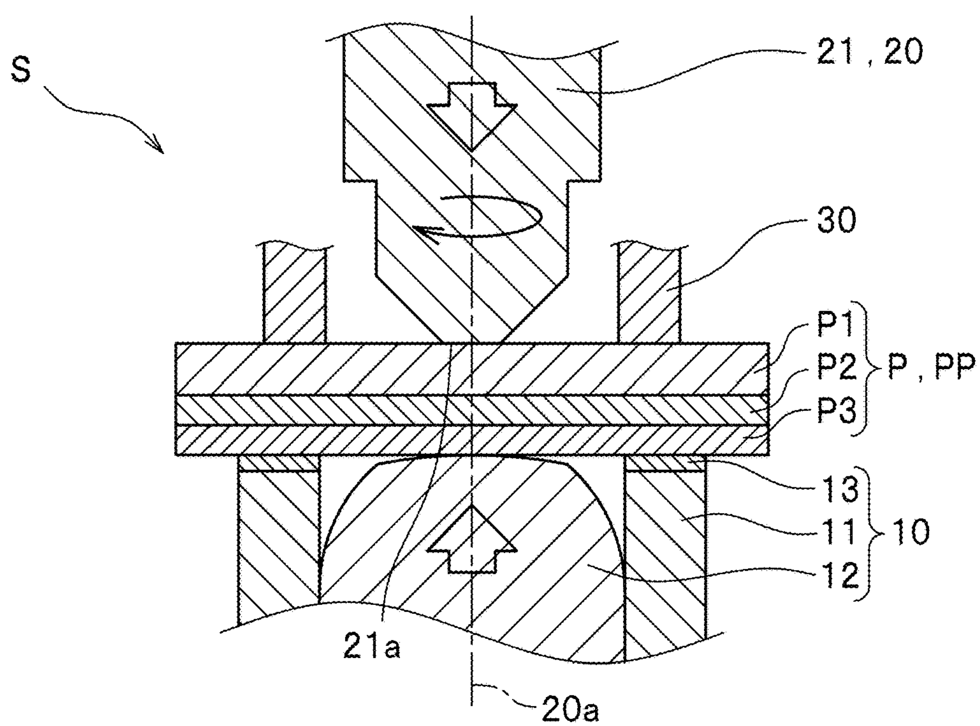

S
21, 20
30
P1
P2
P3
P, PP
13
11
12
10
21a
20a

FIG. 4

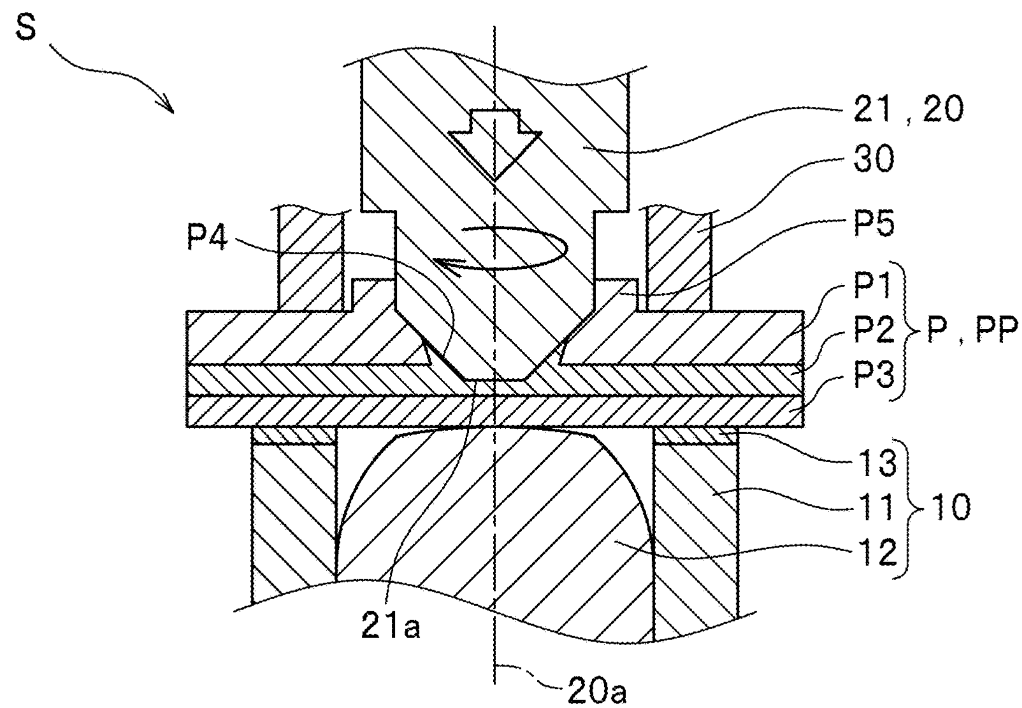

S
21, 20
30
P4
P5
P1
P2
P, PP
P3
13
11
10
12
21a
20a

FIG. 5

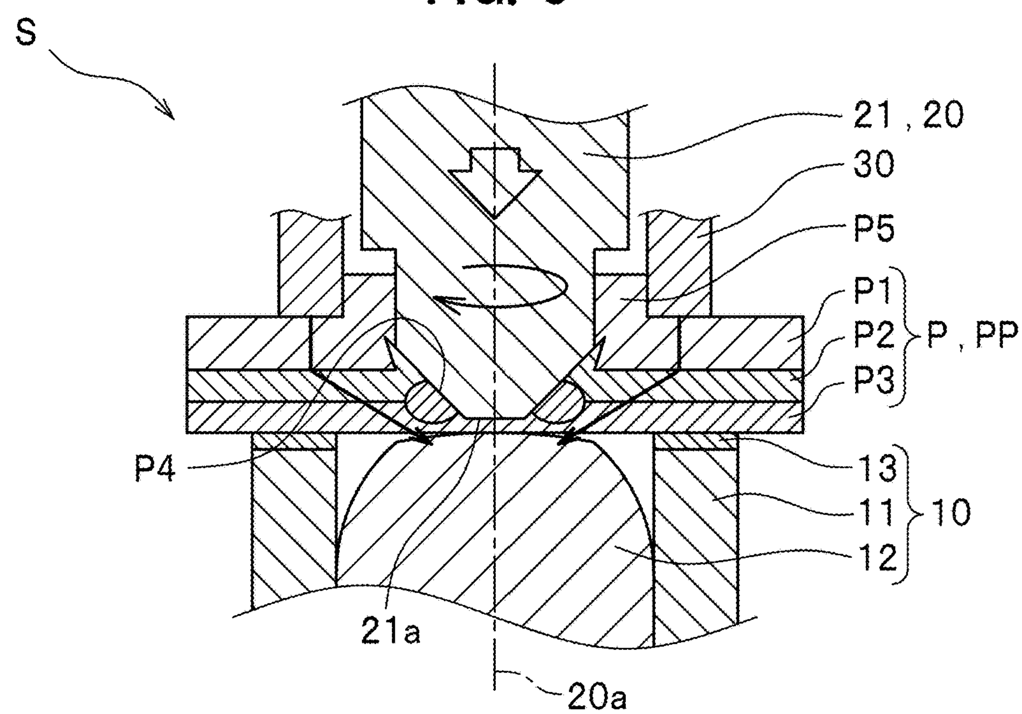

S
21, 20
30
P5
P1
P2
P, PP
P3
P4
13
11
10
12
21a
20a

FIG. 6

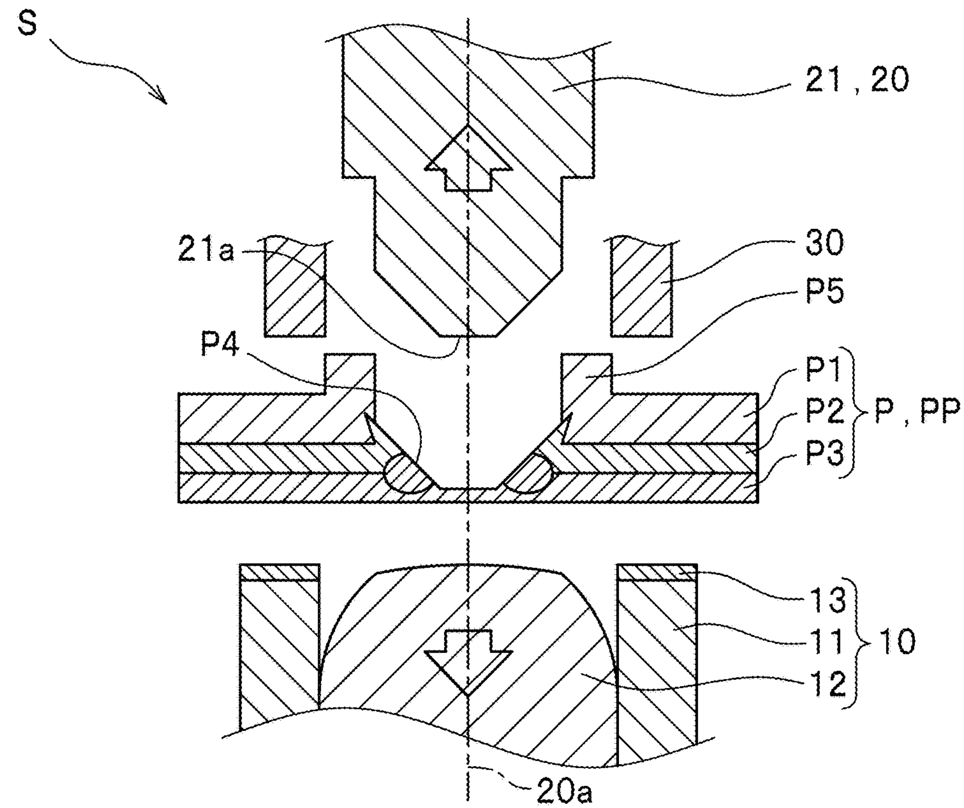

S
21, 20
21a
30
P5
P4
P1
P2
P3
P, PP
13
11
10
12
20a

FIG. 7

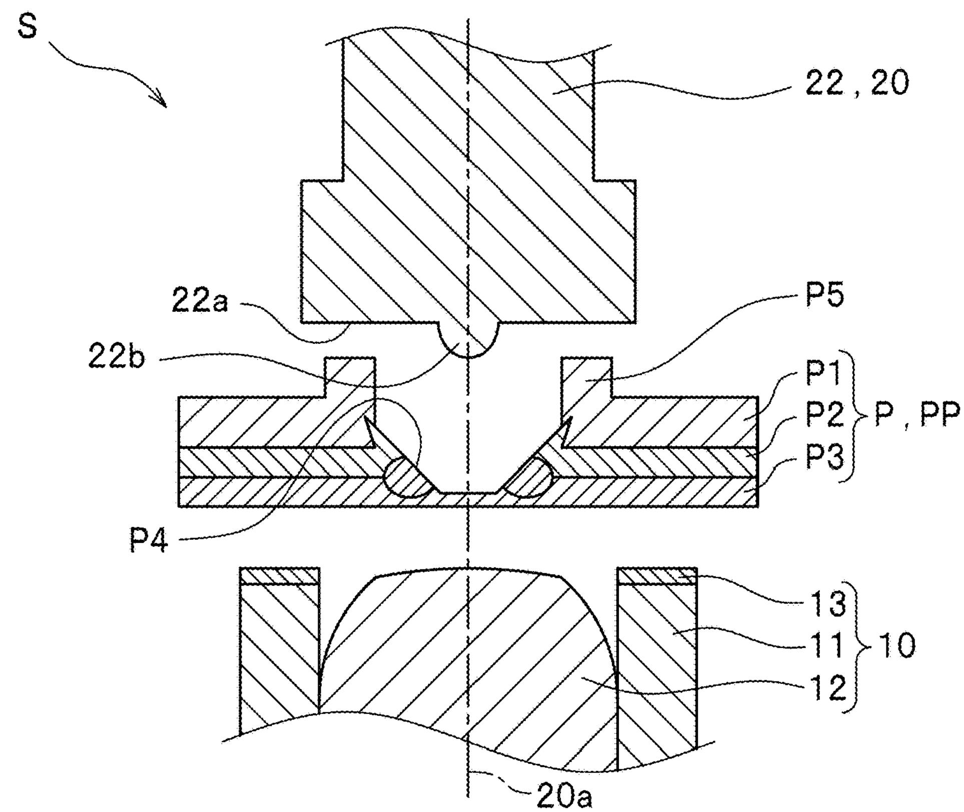

S
22, 20
22a
P5
22b
P1
P2
P3
P, PP
P4
13
11
10
12
20a

FIG. 8

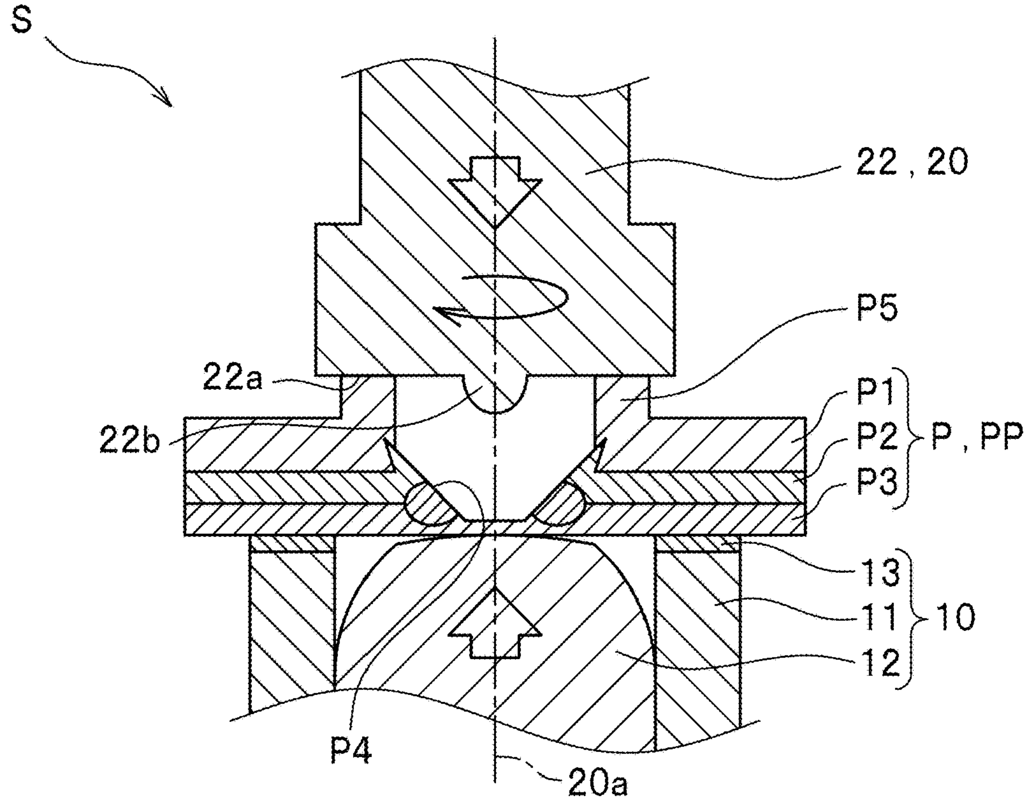

S
22, 20
P5
22a
22b
P1
P2
P3
P, PP
13
11
10
12
P4
20a

FIG. 9

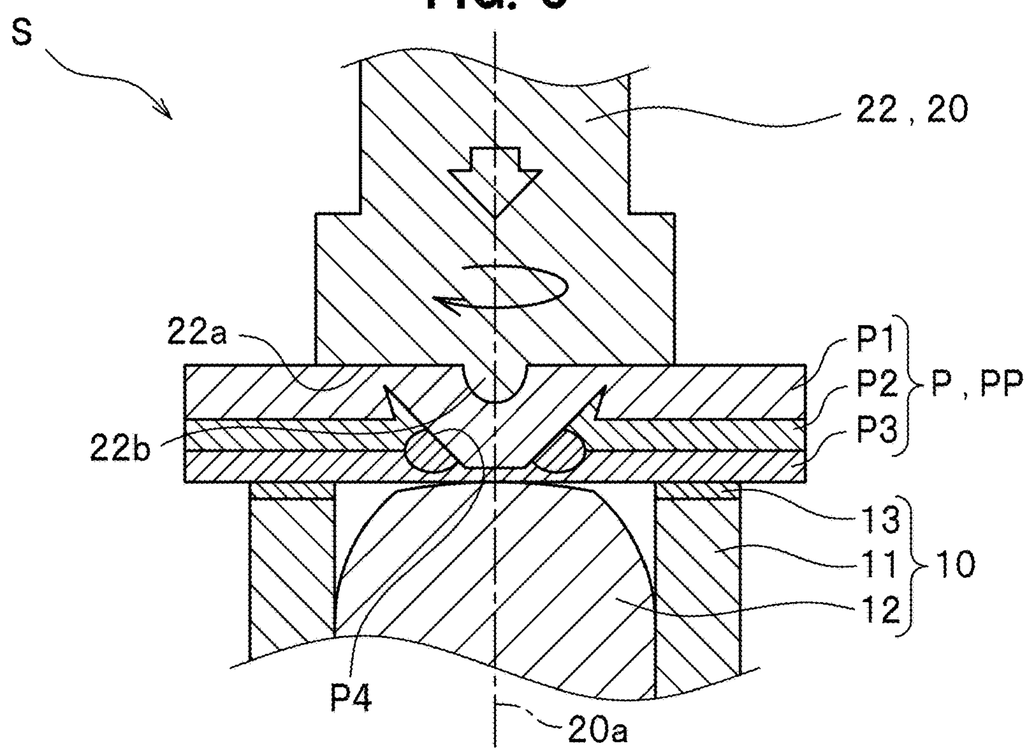

S
22, 20
22a
22b
P1
P2
P3
P, PP
13
11
10
12
P4
20a

FIG. 10

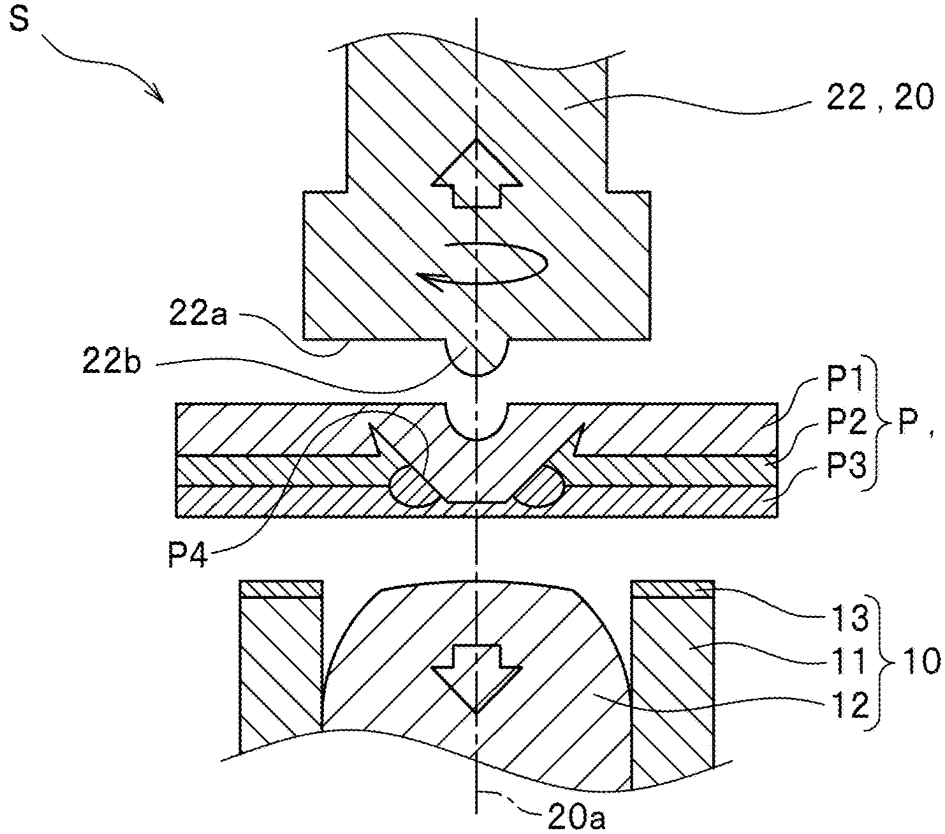

S
22, 20
22a
22b
P1
P2
P3
P, PP
P4
13
11
12
10
20a

FIG. 11

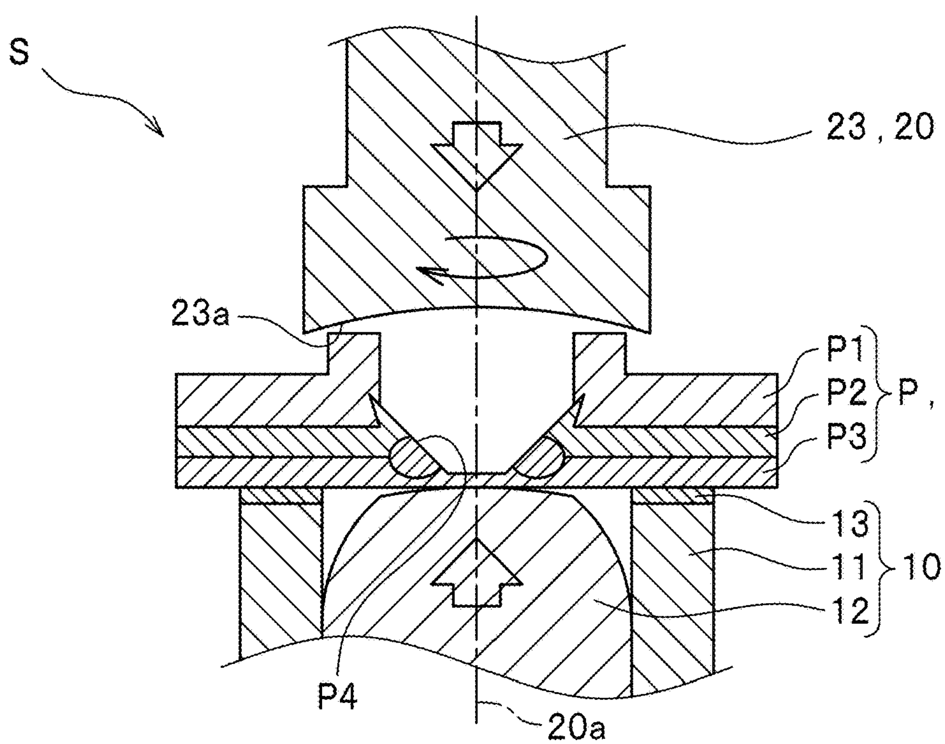

S
23, 20
23a
P1
P2
P3
P, PP
13
11
12
10
P4
20a

FIG. 12

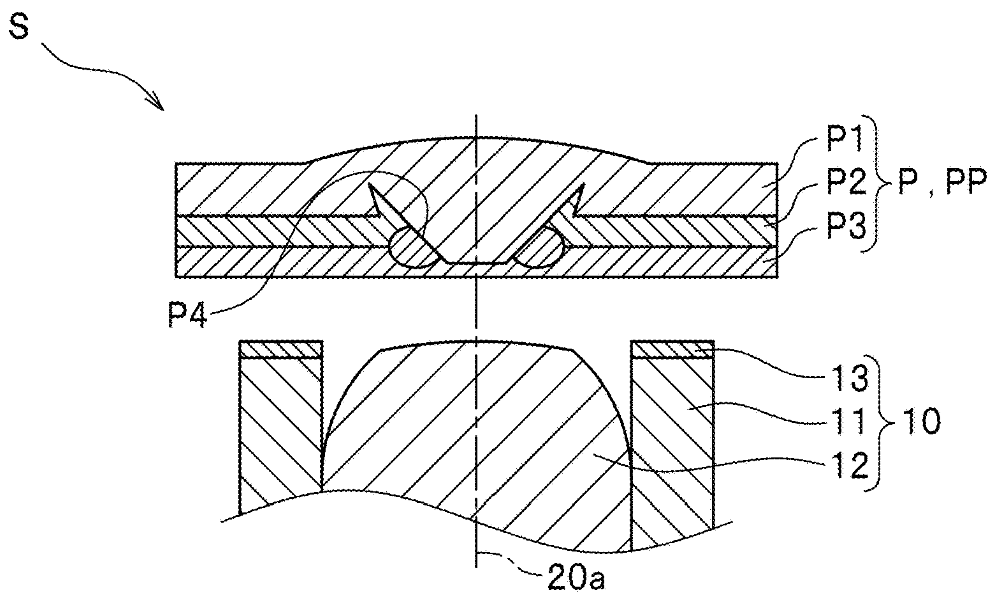

S
P1
P2
P3
P , PP
P4
13
11
10
12
20a

FIG. 13

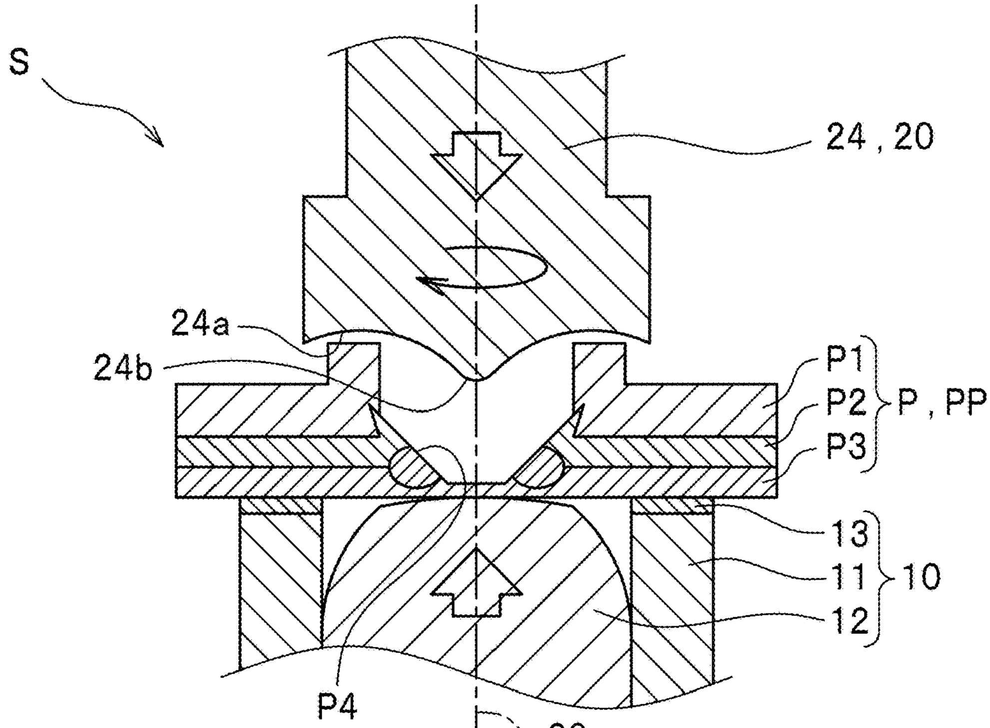

S
24 , 20
24a
24b
P1
P2
P3
P , PP
13
11
10
12
P4
20a

FIG. 14

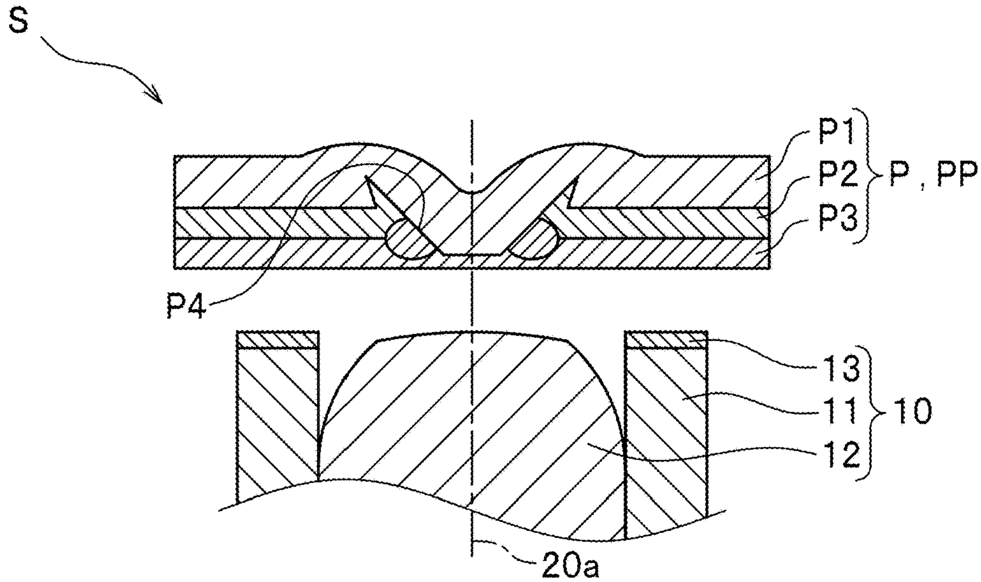


FIG. 15

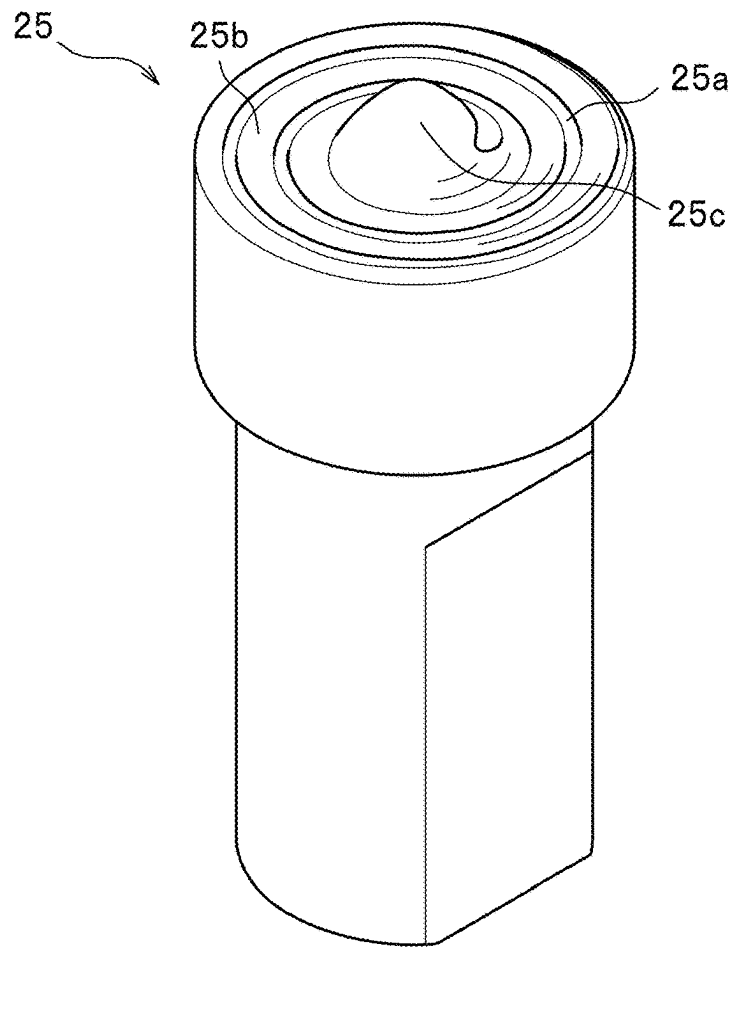


FIG. 16

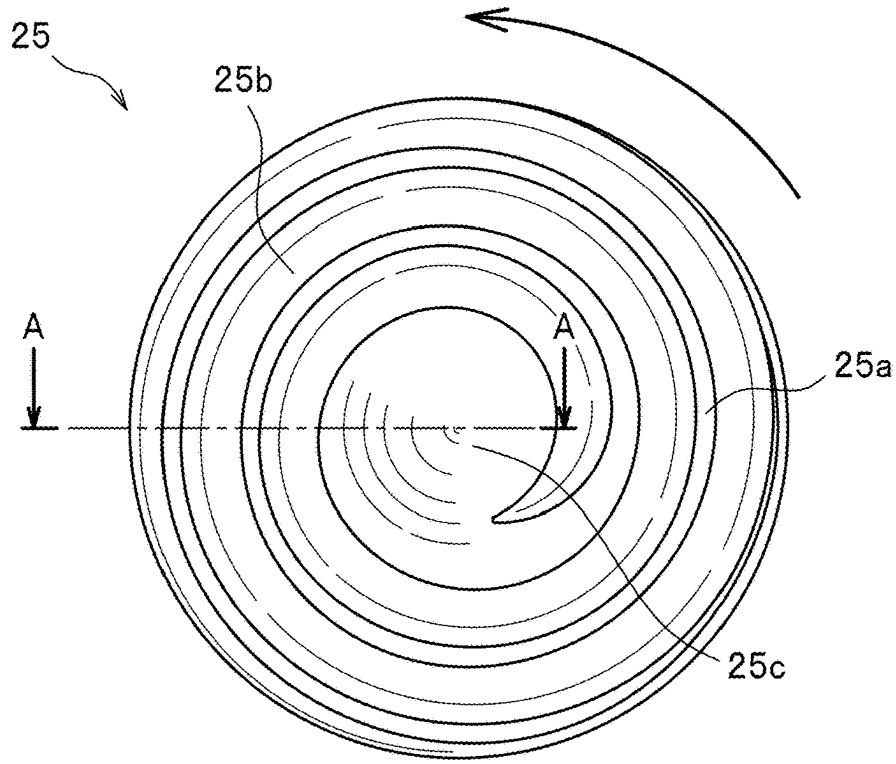


FIG. 17

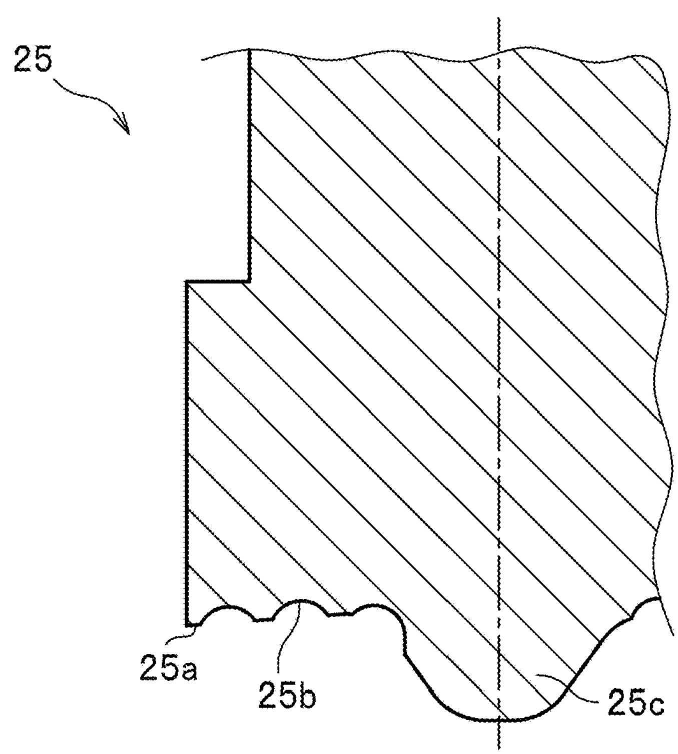
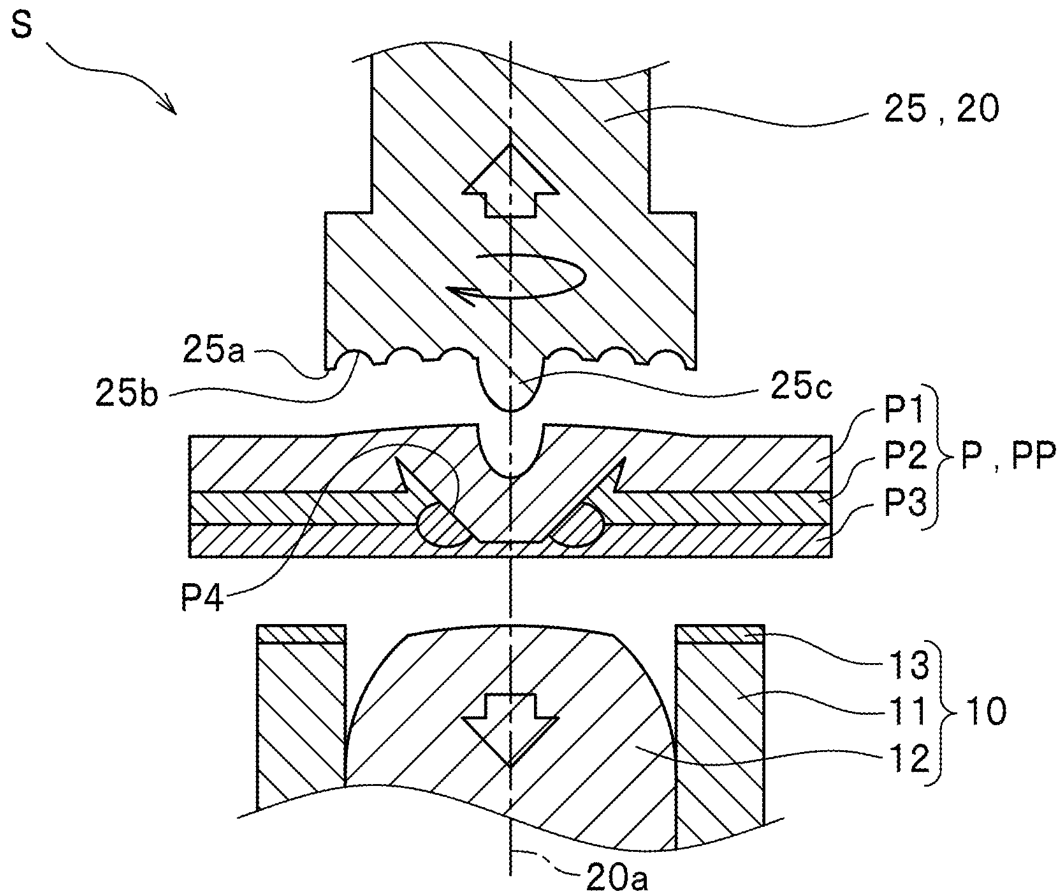


FIG. 18



FRICTION STIR WELDING APPARATUS AND FRICTION STIR WELDING METHOD

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of foreign priority to Japanese Patent Application No. 2024-034655, filed on Mar. 7, 2024, which is incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a friction stir welding apparatus and a friction stir welding method for joining stacked plate members using friction stirring.

2. Description of the Related Art

[0003] In recent years, there have been strong demands for reduction in adverse influence on the environment (SDGs 11.6) and improvement in energy efficiency (SDGs 7.3) regarding the quality of the atmosphere in a production process of products.

[0004] In such trends, the approach of joining a plurality of members has been reviewed.

[0005] For example, as methods for joining a plurality of metal plate members, welding methods such as arc welding have been widely employed. However, a joining method using friction stirring as proposed in JP2023-013804A has been attracting attention because gas generated during the joining step can be reduced and consumed power can be reduced as compared with the above joining methods.

SUMMARY OF THE INVENTION

[0006] Meanwhile, in the case of joining metals having different properties by using friction stirring, a material adjacent portion having a large potential difference is exposed to the surface at a joined portion.

[0007] Hence, in the case where a joining portion comes into contact with water due to condensation or the like, electric corrosion occurs. To prevent such electric corrosion, an anti-corrosion treatment (a chemical conversion treatment, a waterproof sealer application, or the like) is required, causing a problem that steps are complicated.

[0008] The present invention has been made in view of the aforementioned points, and an object of the present invention is to provide a friction stir welding apparatus and a friction stir welding method which can suppress the generation of electric corrosion in a joining step without newly providing a step of applying a waterproof sealer to a material adjacent portion.

[0009] To achieve the above-described object, a friction stir welding apparatus according to the present invention comprises: an anvil which supports a stacked body including stacked plate members; a probe which is disposed to face the anvil, and is disposed to be capable of advancing and retreating relative to the stacked body and capable of rotating about a rotation axis along an advancing and retreating direction; and a shoulder member which has a cylindrical shape through which the probe is capable of being inserted, and holds the stacked body between the shoulder member and the anvil, wherein the probe includes: a joining probe which advances toward the plate members, forms a joint hole in the plate members softened by frictional heat generated by a joining surface of the joining probe coming into sliding contact with the plate members and joins the plate members, and forms an overlay portion protruding in a bank shape from a plate surface around the joint hole along an inner peripheral surface of the shoulder member; and a backfilling probe which advances toward the plate members, and backfills the joint hole with the overlay portion softened by frictional heat generated by a backfilling surface of the backfilling probe coming into sliding contact with the overlay portion, and the joining probe and the backfilling probe are disposed to be replaceable with each other.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present invention in any way.

[0011] FIG. **1** is a schematic configuration diagram illustrating a friction stir welding apparatus according to one embodiment of the present invention.

[0012] FIG. **2** is an enlarged sectional view of a main part, illustrating a state in which a stacked body has been positioned during a joining step by friction stirring.

[0013] FIG. **3** is an enlarged sectional view of a main part, illustrating a state in which a joining probe has advanced and is in sliding contact with the stacked body during the joining step by friction stirring.

[0014] FIG. **4** is an enlarged sectional view of a main part, illustrating a state in which a joint hole is formed and an overlay portion is formed in the stacked body during the joining step by friction stirring.

[0015] FIG. **5** is an enlarged sectional view of a main part, illustrating a state in which plate members have been welded by a resistance welder during the joining step by friction stirring.

[0016] FIG. **6** is an enlarged sectional view of a main part, illustrating a state in which the joining probe has retreated in the joining step by friction stirring.

[0017] FIG. **7** is an enlarged sectional view of a main part, illustrating a state in which the joining probe has been replaced with a backfilling probe in the joining step by friction stirring.

[0018] FIG. **8** is an enlarged sectional view of a main part, illustrating a state in which the backfilling probe has advanced and is in sliding contact with an overlay portion in the joining step by friction stirring.

[0019] FIG. **9** is an enlarged sectional view of a main part, illustrating a state in which the joint hole has been backfilled in the joining step by friction stirring.

[0020] FIG. **10** is an enlarged sectional view of a main part, illustrating a state in which the backfilling probe has retreated and the joining step has been completed in the joining step by friction stirring.

[0021] FIG. **11** is an enlarged sectional view of a main part, illustrating a state in which a backfilling probe of a first modification has advanced and is in sliding contact with an overlay portion in the joining step by friction stirring.

[0022] FIG. **12** is an enlarged sectional view of a main part, illustrating a state in which a joint hole has been backfilled by the first modification in the joining step by friction stirring.

[0023] FIG. **13** is an enlarged sectional view of a main part, illustrating a state in which a backfilling probe of a second modification has advanced and is in sliding contact with an overlay portion in the joining step by friction stirring.

[0024] FIG. **14** is an enlarged sectional view of a main part, illustrating a state in which a joint hole has been backfilled by the second modification in the joining step by friction stirring.

[0025] FIG. **15** is a perspective view showing a backfilling probe of a third modification.

[0026] FIG. **16** is a plan view showing a backfilling surface in the backfilling probe of the third modification.

[0027] FIG. **17** is a sectional view taken along line A-A in FIG. **16**.

[0028] FIG. **18** is an enlarged sectional view of a main part, illustrating a state in which a joint hole has been backfilled by the third modification in the joining step by friction stirring.

DETAILED DESCRIPTION

[0029] A friction stir welding apparatus S according to one embodiment of the present invention will be described in detail with reference to FIGS. **1** to **10**.

[0030] Note that in the description, the same elements are denoted by the same reference signs, and repetitive descriptions will be omitted.

[0031] The friction stir welding apparatus S of the present embodiment joins plate members P stacked in a plate thickness direction with each other (see FIG. **1**).

[0032] Then, the stacked plate members P will be referred to as a stacked body PP.

[0033] Note that in the friction stir welding apparatus S of the present embodiment, the stacked body PP is disposed such that its plate surfaces face in upper-lower directions.

[0034] The stacked body PP is configured with stacked three plate members P made of metals (see FIG. **1**).

[0035] In the stacked body PP, a plate member P (an upper member P**1**) that is located at the top is formed of an aluminum alloy.

[0036] In addition, in the stacked body PP, a plate member P (a middle member P**2**) that is located in the middle and a plate member P (a lower member P**3**) that is located at the bottom are formed of the same iron alloy.

[0037] That is, the upper member P**1** is formed of a material having a lower melting point than that of the middle member P**2** and the lower member P**3**.

[0038] The friction stir welding apparatus S includes an anvil **10**, a probe **20**, a shoulder member **30**, a drive mechanism **40**, a resistance welder **50**, and a controller **60**.

[0039] The anvil **10** is disposed to be capable of advancing and retreating relative to the probe **20** (the stacked body PP) along a rotation axis **20***a* of the probe **20**, and is moved by the drive mechanism **40**.

[0040] Note that the direction along the rotation axis **20***a* is referred to as an advancing and retreating direction.

[0041] The anvil **10** supports the stacked body PP from below in the state of having advanced toward the probe **20**.

[0042] The anvil **10** includes a receiving portion **11** and a semi-spherical portion **12**.

[0043] The receiving portion **11** and the semi-spherical portion **12** are formed of separate members.

[0044] The receiving portion **11** has a cylindrical shape, and is disposed to be capable of moving along the rotation axis **20***a*.

[0045] The receiving portion **11** includes an insulating material **13** disposed on an upper end surface thereof which comes into contact with the stacked body PP.

[0046] The semi-spherical portion **12** is formed with a column-shaped member having a semi-spherical shape protruding upward in an upper end portion thereof, and is disposed inside the cylinder of the receiving portion **11**.

[0047] The semi-spherical portion **12** is disposed to be capable of moving along the rotation axis **20***a* together with the receiving portion **11**.

[0048] The semi-spherical portion **12** is formed of a conductive material.

[0049] The anvil **10** is configured such that a front end of the semi-spherical portion **12** comes into contact with the stacked body PP while the receiving portion **11** supports the stacked body PP.

[0050] The probe **20** joins and backfills the stacked body PP.

[0051] The probe **20** is disposed to be capable of rotating about the rotation axis **20***a* and capable of advancing and retreating relative to the anvil **10** (the stacked body PP) along the rotation axis **20***a*.

[0052] The probe **20** rotates and moves with the drive mechanism **40** as a drive source.

[0053] The probe **20** is formed of a material such as ceramic or cemented carbide.

[0054] That is, for the probe **20**, a material that is harder and has a higher melting point than the material of the stacked body PP is selected.

[0055] The probe **20** includes a joining probe **21** and a first backfilling probe **22** (a backfilling probe).

[0056] The probe **20** is configured to be attachable to and detachable from the friction stir welding apparatus S and allow the joining probe **21** and the first backfilling probe **22** to be replaced with each other.

[0057] The joining probe **21** joins three plate members P included in the stacked body PP with each other.

[0058] The joining probe **21** has a substantially columnar shape and has a tapered front end portion having a tapered shape.

[0059] The joining probe **21**, while rotating, advances toward the stacked body PP from an upper side to a lower side in FIG. **1** to FIG. **10**, and a joining surface **21***a* formed in the front end thereof comes into sliding contact with the stacked body PP.

[0060] The sliding contact generates frictional heat between the front end of the joining probe **21** and the stacked body PP, and the frictional heat softens the stacked body PP in the order of the upper member P**1**, the middle member P**2**, and the lower member P**3**.

[0061] Then, as the joining probe **21** further advances, the front end of the joining probe **21** pushes each member of the softened stacked body PP aside, forming a joint hole P**4** and joining the members with each other.

[0062] In addition, an overlay portion P**5** protruding in a bank shape from a plate surface is formed inside an inner peripheral surface of the shoulder member **30** and around the joint hole P**4** by the members pushed aside.

[0063] The first backfilling probe **22** backfills the joint hole P**4** with the overlay portion P**5**.

[0064] The first backfilling probe **22** has a substantially columnar shape and is provided, on a front end thereof, with a first backfilling surface **22***a* (a backfilling surface) formed of a circular flat surface which is orthogonal to the rotation axis **20***a* and has a diameter larger than the outer diameter of the overlay portion P**5** (an inner diameter of the shoulder member **30**).

[0065] Then, a first backfill protrusion **22***b* is formed on the center of the first backfilling surface **22***a* (on the rotation axis **20***a*).

[0066] The first backfill protrusion **22***b* has a substantially conical shape, and protrudes toward the stacked body PP.

[0067] Note that the protrusion dimension of the first backfill protrusion **22***b* from the first backfilling surface **22***a* is set to be smaller than the plate-thickness dimension of the upper member P**1**, and the diameter of the base portion of the first backfill protrusion **22***b* is set to be smaller than the hole diameter of the joint hole P**4**.

[0068] That is, the shape of the first backfill protrusion **22***b* is set such that layers of aluminum alloy inside the joint hole P**4** after the backfilling have dimensions sufficient to suppress generation of electric corrosion.

[0069] The shoulder member **30** has a circular cylindrical shape about the rotation axis **20***a* as a central axis.

[0070] Inside the cylinder of the shoulder member **30**, the probe **20** rotates, and advances and retreats.

[0071] The shoulder member **30** is disposed to be capable of advancing and retreating relative to the anvil **10** (the stacked body PP) along the rotation axis **20***a*.

[0072] The shoulder member **30** is caused to advance and retreat, and move by a shoulder drive mechanism **44** as a drive source.

[0073] The shoulder member **30** is formed of a conductive material.

[0074] The drive mechanism **40** includes a rotating drive mechanism **41**, an advancing and retreating drive mechanism **42**, an anvil drive mechanism **43**, and the shoulder drive mechanism **44**.

[0075] The rotating drive mechanism **41** is a drive source for causing the probe **20** to rotate.

[0076] The advancing and retreating drive mechanism **42** is a drive source for causing the probe **20** to advance and retreat.

[0077] The anvil drive mechanism **43** is a drive source for causing the anvil **10** to advance and retreat.

[0078] The shoulder drive mechanism **44** is a drive source for causing the shoulder member **30** to advance and retreat.

[0079] The resistance welder **50** causes current to flow through the stacked body PP by using the semi-spherical portion **12** of the anvil **10** and the shoulder member **30** as electrodes to weld members included in the stacked body PP with each other.

[0080] The controller **60** controls the drive mechanism **40** and the resistance welder **50**.

[0081] Next, an operation of the friction stir welding apparatus S (joining procedure) will be described (see FIGS. **2**-**10**).

[0082] First, a step of causing the anvil **10** to support the stacked body PP is conducted (see FIG. **2**).

[0083] Note that before this step is conducted, the joining probe **21** is attached to the apparatus body (see FIG. **2**).

[0084] In the present step, first, the stacked body PP is placed on the anvil **10**.

[0085] Here, the stacked body PP is disposed such that a portion to be joined is located on the semi-spherical portion **12**.

[0086] Then, the shoulder member **30** is caused to advance to hold and fix the stacked body PP between the shoulder member **30** and the anvil **10**.

[0087] Note that although the procedure of moving the stacked body PP in conformity to the position of the friction stir welding apparatus S has been described above, the procedure is not limited to this.

[0088] For example, a procedure of moving and changing the position and direction of the friction stir welding apparatus S in conformity to the stacked body PP without moving the stacked body PP from the original position may be employed.

[0089] Next, a step of pressing the joining surface **21***a* of the joining probe **21** against the stacked body PP, and softening the stacked body PP with frictional heat generated by sliding contact to join the plate members P each other and form an overlay portion P**5** is conducted (see FIGS. **3** to **6**).

[0090] In the present step, the controller **60** activates the drive mechanism **40** to cause the joining probe **21** to rotate and advance toward the stacked body PP (see FIG. **3**).

[0091] Then, the rotating joining surface **21***a* is pressed against the stacked body PP to sequentially soften each plate member P with frictional heat generated by sliding contact.

[0092] Moreover, the joint hole P**4** is formed by causing the joining probe **21** to rotate and advance toward the stacked body PP, so that the joining probe **21** pushes the softened members aside (see FIG. **4**).

[0093] Then, the overlay portion P**5** is formed around the joint hole P**4** by the members pushed aside, and the plate members P are joined with each other.

[0094] In addition, once the dimension of advancement of the joining probe **21** reaches a set value, the controller **60** activates the resistance welder **50** to weld the middle member P**2** and the lower member P**3** (see FIG. **5**).

[0095] After the welding, the controller **60** stops the resistance welder **50**, and causes the joining probe **21** to rotate and retreat, separating the joining probe **21** and the anvil **10** from the stacked body PP (see FIG. **6**).

[0096] Note that a hole wall of the joint hole P**4** in this state is left exposed as a material adjacent portion having a large potential difference.

[0097] Next, a step of replacing the joining probe **21** with the first backfilling probe **22** is conducted (see FIG. **7**).

[0098] In the present step, the probe **20** is replaced by detaching the joining probe **21** from the apparatus body (not shown) of the friction stir welding apparatus S, and attaching the first backfilling probe **22** instead.

[0099] Note that the shoulder member **30** may be moved to a position which does not interfere with the first backfilling probe **22**.

[0100] Next, a step of pressing the first backfilling surface **22***a* of the first backfilling probe **22** against the overlay portion P**5**, softening the overlay portion P**5** with frictional heat generated by sliding contact, and backfilling the joint hole P**4** with the softened overlay portion P**5** by using the first backfilling surface **22***a* is conducted (see FIGS. **8** to **10**).

[0101] In the present step, the first backfilling probe **22** is caused to rotate and advance toward the overlay portion P**5** (the stacked body PP) (see FIG. **8**).

[0102] The first backfilling surface **22***a* being rotated is pressed against the overlay portion P**5** to soften the overlay portion P**5** with frictional heat generated by sliding contact.

[0103] The first backfilling surface **22***a* is caused to further advance to backfill the joint hole P**4** with the softened portion of the overlay portion P**5** (see FIG. **9**).

[0104] Then, as the joint hole P**4** is backfilled, the material adjacent portion of the exposed hole wall is covered up.

[0105] Once the dimension of advancement of the first backfilling probe **22** reaches a set value, the controller **60** causes the joining probe **21** to rotate and retreat, separating the first backfilling probe **22** and the anvil **10** from the stacked body PP (see FIG. **6**).

[0106] The joining step in one joining portion is completed in the above-described manner, and then the joining step in the next joining portion is executed.

[0107] Next, advantageous effects of the present embodiment will be described.

[0108] By using the friction stir welding apparatus S of the present embodiment, the joint hole P**4** which is generated at the time of joining can be backfilled in the joining step.

[0109] Then, since the material adjacent portion having a large potential difference can be covered up without being exposed on the surface, the surface of the materials can be protected from electric corrosion.

[0110] This eliminates the need of newly providing a step of applying a waterproof sealer to the joining surface for preventing electric corrosion.

[0111] In addition, backfilling the joint hole P**4** increases the broken area, the joint strength between the lower member P**3** and the middle member P**2** can be improved.

[0112] According to the friction stir welding apparatus S and the friction stir welding method of the present embodiment, it is possible to conduct friction stir welding which can suppress the generation of electric corrosion in a joining step without newly providing a step of applying a waterproof sealer to a material adjacent portion.

[0113] In addition, in the friction stir welding apparatus S of the present embodiment, the first backfill protrusion **22***b* is provided on the center of the first backfilling surface **22***a* of the first backfilling probe **22**.

[0114] When the joint hole P**4** is backfilled with the overlay portion P**5**, the joint hole P**4** is backfilled gradually from the outer peripheral side of the hole.

[0115] However, because no friction is generated between the softened members and the first backfilling surface **22***a*, there is a possibility that the softened members are cooled and hardened during moving to the central side of the hole, causing a filling failure.

[0116] In view of this, the friction surface near the center is increased by providing the first backfill protrusion **22***b*.

[0117] With this, the first backfill protrusion **22***b* can come into sliding contact with members which are hardened in the course of moving from the outer peripheral side to the central side of the hole, and re-soften the members being hardened, so that the portion around the center can be surely backfilled.

[0118] Note that in the friction stir welding apparatus S of the present embodiment, the joining probe **21** and the first backfilling probe **22** are attached to and detached from the apparatus body for replacement; however, the configuration is not limited to this design.

[0119] For example, the joining probe **21** and the first backfilling probe **22** may be arranged adjacent to the apparatus body, enabling the probe **20** to perform lateral or rotational movements during joining and backfilling.

[0120] That is, the joining probe **21** and the first backfilling probe **22** can be replaced without removing them from the apparatus body. The same advantageous effects as described above can be obtained by this configuration.

[0121] When a plurality of joining portions are set on the stacked body PP, a procedure can be used where all the joining portions are first joined using the joining probe **21**. Then, the probe **20** is replaced with the first backfilling probe **22**, and all the joint holes P**4** are backfilled.

[0122] This can reduce the time taken for the replacement of the probe, and the working time taken for joining and backfilling can thus be shortened.

[0123] Moreover, a unit (a joining unit) consisting of the joining probe **21** and the anvil **10** and a unit (a backfilling unit) consisting of the first backfilling probe **22** and the anvil **10** can be configured separately in a single friction stir welding apparatus S.

[0124] Then, by arranging these units adjacent to each other and replacing the probe **20** through unit replacement, the same advantageous effects as described above can be obtained.

[0125] In addition, in the case where such a configuration is employed, while backfilling is conducted by the backfilling unit, joining of another portion can be simultaneously conducted by the joining unit.

[0126] Hence, such a configuration is more favorable because the working time can be shortened in the case where a plurality of portions have to be joined.

[0127] According to the friction stir welding apparatus S of the present embodiment, the stacked body PP is stacked one on top of another, and the probe **20** joins or backfills the stacked body PP from above while the anvil **10** supports the stacked body PP from below. However, the friction stir welding apparatus S is not limited to this configuration.

[0128] For example, the stacked body PP may be stacked in a lateral direction, and the probe **20** may join or backfill the stacked body PP from the left side while the anvil **10** supports the stacked body PP from the right side.

[0129] This means that joining and backfilling can be conducted regardless of the direction in which plate members are stacked.

[0130] Next, a first modification of the backfilling probe will be described in detail with reference to FIGS. **11** and **12**.

[0131] Note that in the description, the same elements as in the aforementioned embodiment are denoted by the same reference signs, and repetitive descriptions will be omitted.

[0132] In a second backfilling probe **23** of this modification, the shape of a second backfilling surface **23***a* is different from that of the first backfilling surface **22***a* of the aforementioned embodiment.

[0133] The second backfilling surface **23***a* of the second backfilling probe **23** of the present modification has a circular shape which is formed about a rotation axis **20***a* such that an outer peripheral edge of the circular shape is located outside an outer peripheral edge of an overlay portion P**5**.

[0134] In addition, the second backfilling surface **23***a* has a concave surface shape formed of a curved surface which is depressed such that a center portion of the second backfilling surface **23***a* is farthest away from a plate surface of a stacked body PP.

[0135] That is, the outer peripheral edge of the second backfilling surface **23***a* is provided at a position farther away from the rotation axis **20***a* than an inner periphery of a shoulder member **30** is.

[0136] In addition, the second backfilling surface **23***a* has a concave surface shape which is depressed rearward in the advancing direction.

[0137] By using the second backfilling probe **23** of the present modification to backfill a joint hole P**4**, softened members are pressed from the outer peripheral side of the second backfilling surface **23***a* toward the central side thereof.

[0138] This inhibits flowing out of the softened members to the outer side of the second backfilling surface **23***a,* and thus makes it possible to more surely backfill the joint hole P**4**.

[0139] Next, a second modification of the backfilling probe will be described in detail with reference to FIGS. **13** and **14**.

[0140] Note that in the description, the same elements as in the aforementioned embodiment are denoted by the same reference signs, and repetitive descriptions will be omitted.

[0141] In a third backfilling probe **24** of this modification, the shape of a third backfilling surface **24***a* is different from that of the first backfilling surface **22***a* of the aforementioned embodiment.

[0142] The third backfilling surface **24***a* of the third backfilling probe **24** of the present modification has a shape in which the first backfill protrusion **22***b* of the first backfilling probe **22** of the above embodiment and the second backfilling surface **23***a* of the second backfilling probe **23** of the first modification are combined.

[0143] That is, a third backfill protrusion **24***b* having a substantially conical shape is provided on the center of a concave surface shape.

[0144] With such a configuration, softened members can be moved from the outer peripheral side to the central side with the concave surface shape portion, and the third backfill protrusion **24***b* can come into sliding contact with the members which are cooled and hardened in the course of moving to the central side to re-soften the members.

[0145] This makes it possible to more surely backfill the center portion of the joint hole P**4**.

[0146] Next, a third modification of the backfilling probe will be described in detail with reference to FIGS. **15** to **18**.

[0147] Note that in the description, the same elements as in the aforementioned embodiment are denoted by the same reference signs, and repetitive descriptions will be omitted.

[0148] In a fourth backfilling probe **25** of this modification, the shape of a fourth backfilling surface **25***a* is different from that of the first backfilling surface **22***a* of the aforementioned embodiment.

[0149] The fourth backfilling surface **25***a* of the fourth backfilling probe **25** of the present modification has a circular shape which is formed about a rotation axis **20***a* such that an outer peripheral edge of the circular shape is located outside an outer peripheral edge of an overlay portion P**5**.

[0150] In addition, in the fourth backfilling surface **25***a,* a groove which has a spiral shape continuing from the outer peripheral edge to the center (a spiral groove **25***b*) is formed (see FIG. **16**).

[0151] Moreover, in the center of the fourth backfilling surface **25***a,* a fourth backfill protrusion **25***c* which is similar to the first backfill protrusion **22***b* of the above embodiment is provided.

[0152] Then, an end portion of the spiral groove on the central side is connected to the fourth backfill protrusion **25***c.*

[0153] Hence, by rotating the fourth backfilling probe **25** in a counterclockwise direction in FIG. **16**, action of collecting softened members toward the fourth backfill protrusion **25***c* is generated.

[0154] Then, the fourth backfill protrusion **25***c* can come into sliding contact with the members which are cooled and hardened in the course of moving to the central side of the spiral groove **25***b* to re-soften the members.

[0155] By using the fourth backfilling probe **25** of the present modification to backfill a joint hole P**4**, softened members are moved in the groove of the spiral groove **25***b* and pressed from the outer peripheral side of fourth backfilling surface **25***a* to the central side thereof.

[0156] This inhibits flowing out of the softened members to the outer side of the fourth backfilling surface **25***a,* and thus makes it possible to more surely backfill the joint hole P**4**.

What is claimed is:

1. A friction stir welding apparatus comprising:
- an anvil which supports a stacked body including stacked plate members;
- a probe which is disposed to face the anvil, and is disposed to be capable of advancing and retreating relative to the stacked body and capable of rotating about a rotation axis along an advancing and retreating direction; and
- a shoulder member which has a cylindrical shape through which the probe is capable of being inserted, and holds the stacked body between the shoulder member and the anvil, wherein
- the probe includes:
  - a joining probe which advances toward the plate members, forms a joint hole in the plate members softened by frictional heat generated by a joining surface of the joining probe coming into sliding contact with the plate members and joins the plate members, and forms an overlay portion protruding in a bank shape from a plate surface around the joint hole along an inner peripheral surface of the shoulder member; and
  - a backfilling probe which advances toward the plate members, and backfills the joint hole with the overlay portion softened by frictional heat generated by a backfilling surface of the backfilling probe coming into sliding contact with the overlay portion, and
- the joining probe and the backfilling probe are disposed to be replaceable with each other.

2. The friction stir welding apparatus according to claim **1**, wherein
- the backfilling probe is disposed to be rotatable about a rotation axis along the advancing and retreating direction, and
- the backfilling surface has a circular shape about the rotation axis, and includes a backfill protrusion which protrudes toward the stacked body in a center of the backfilling surface.

3. The friction stir welding apparatus according to claim **1**, wherein the backfilling probe is disposed to be rotatable about a rotation axis along the advancing and retreating direction, and

the backfilling surface has a circular shape which is formed about the rotation axis such that an outer peripheral edge of the circular shape is located outside an outer peripheral edge of the overlay portion, and has a concave surface shape which is formed of a curved surface which is depressed such that a center portion of the concave surface shape is farthest away from a plate surface of the stacked body.

4. The friction stir welding apparatus according to claim **1**, wherein

the backfilling probe is disposed to be rotatable about a rotation axis along the advancing and retreating direction, and

the backfilling surface has a circular shape which is formed about the rotation axis such that an outer peripheral edge of the circular shape is located outside an outer peripheral edge of the overlay portion, and has a spiral groove which has a spiral shape continuing from the outer peripheral edge of the circular shape to the rotation axis.

5. A friction stir welding method comprising the steps of:

causing an anvil to support a stacked body;

pressing a joining surface of a joining probe against the stacked body, softening the stacked body with frictional heat generated by sliding contact to join plate members each other and form a joint hole therein and to form an overlay portion;

replacing the joining probe with a backfilling probe; and

pressing a backfilling surface of the backfilling probe against the overlay portion, softening the overlay portion with frictional heat generated by sliding contact, and backfilling the joint hole with the softened overlay portion by using the backfilling surface.

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